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Display assemblies with condensation mitigation and related systems and methods

Abstract

Display assemblies and related systems and methods are provided which provide condensation mitigation. A cover layer forms a forward portion of a housing for an electronic display located rearward of the cover layer. A continuous airflow pathway extends entirely within the housing such that the continuous airflow pathway is partitioned from an ambient environment. One or more gaskets are provided which, at least in part, partition the continuous airflow pathway from the ambient environment. The one or more gaskets extending from an interior surface of the cover layer to a portion of the housing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 18/213,709 filed Jun. 23, 2023, which is a continuation-in-part of U.S. application Ser. No. 17/694,261 filed Mar. 14, 2022, now U.S. Pat. No. 12,022,635 issued Jun. 25, 2024, which claims the benefit of U.S. provisional application Ser. No. 63/161,147 filed Mar. 15, 2021, and also claims the benefit of U.S. provisional application Ser. No. 63/239,273 filed Aug. 31, 2021, the disclosures of each of the foregoing are hereby incorporated by reference as if fully restated herein.

TECHNICAL FIELD

(1) Exemplary embodiments relate generally to display assemblies with condensation mitigation features and related systems and methods.

BACKGROUND AND SUMMARY OF THE INVENTION

(2) The use of electronic displays, such as for advertising, in the out-of-home market has increased in popularity over recent years. Being located outdoors, such electronic displays are frequently exposed to harsh conditions, including, but not limited to, solar loading, extreme temperatures, precipitation, moisture, contaminants, vandalism, wildlife, and the like. To protect the electronic displays and associated sensitive components from such harsh conditions, it is known to place the electronic displays in ruggedized housings. Such housings may fully or partially seal the electronic displays and other associated sensitive components.

(3) It is known to thermally manage such electronic display assemblies using ambient air and/or circulating gas. Such ambient air may pass through one or more open loop airflow pathways within the assembly, and may thermally interact with circulating gas in one or more closed loop airflow pathways within the assembly where such closed loop pathways are used.

(4) Operating such display assemblies in certain environments may result in the introduction of ambient air, such as into the one or more open loop airflow pathways, having a sufficiently different temperature relative to circulating gas and/or components of the display assembly as to result in the formation of condensation inside such display assemblies. For example, without limitation, the introduction of relatively cool ambient air into the display assembly may result in a sufficiently low dewpoint within the display assembly that water vapor in the ambient air and/or circulating gas within the assembly condenses into liquid, which may cause fogging and/or undesirable moisture exposure to sensitive electronic components. More specifically, for example, without limitation, the introduction of relatively cool ambient air into air-to-air heat exchangers contained within the display assembly may cause surfaces of these heat exchangers to drop below the dewpoint of the relative humidity contained within the fully or partially sealed enclosure resulting in condensation. Additionally, any inside surface of the fully or partially sealed enclosure may reach, or even drop below, the outside ambient temperature (e.g., when ice/snow is piled on top of a housing of the assembly), resulting in cold spots within the assembly whose temperature is below the internal dewpoint which may cause the formation of condensation.

(5) Furthermore, owners, operators, and/or manufacturers of such electronic display assemblies are increasingly undertaking power efficiency efforts. Such power efficiency efforts may include, for example, without limitation, decreasing illumination levels for lighting elements of the electronic display assemblies at night (e.g., partially or to zero), which may result in the electronic display assemblies becoming relatively cool, increasing the likelihood of condensation by lowering the dewpoint inside the display assembly. This may be particularly prevalent when combined with the ingestion of relatively cool ambient air.

(6) What is needed are systems and methods for controlling condensation in electronic display assemblies. Systems and methods for controlling condensation in electronic display assemblies are provided.

(7) In general, gaskets utilized in such electronic display assemblies may be sufficient to entirely or substantially keep out liquids, but sometimes such gaskets are not gas-tight or entirely gas-tight. Therefore, moisture can sometimes permeate through the gasket, such as in the form of water vapor, and enter an otherwise closed loop airflow pathway. Such closed loop airflow pathways may still be considered sealed, as such closed loop airflow pathways are kept entirely or substantially free from solid or liquid particulate such as, but not limited to, dust, debris, precipitation, combinations thereof, or the like. In general, when the interior of an otherwise fully or partially sealed enclosure (e.g., closed loop airflow pathway) is warmer than the outdoor environment, moisture may escape the otherwise fully or partially sealed enclosure, such as, but not limited to, by way of gaseous particles in the air which may permeate the liquid-tight, but not necessarily vapor-tight, gaskets. More specifically, for example, without limitation, as heat is added to the circulating gas within the closed loop airflow pathways, the circulating gas may expand and some of that expanding air may be forced through the gasket or otherwise to a location outside of the closed loop airflow pathway, and take moisture in the circulating gas with it. The converse may also be true.

(8) The relative humidity within an electronic display assembly may be determined, such as, but not limited to, by way of one or more sensors configured to measure humidity and/or temperature. The relative humidity may be determined at the one or more sensors or at a separate controller. The same or different temperature measures may be used in conjunction with the relative humidity to determine the dewpoint of air inside the electronic display assembly. The measurements from multiple sensors may be used and/or multiple readings may be aggregated in various ways.

(9) A dewpoint spread may be calculated between the temperature of ambient air and the calculated dewpoint. For example, without limitation, the temperature of the ambient air, such as for calculating the dewpoint and/or dewpoint spread, may be determined as the lesser of one or more temperature readings of ambient air at said intake or along said portion of said one or more open loop airflow pathways of the assembly or an internet-retrieved local ambient air temperature.

(10) Where the dewpoint spread is determined to be less than 0°C ., it may be determined with a high degree of confidence that condensation is occurring. Where the dewpoint spread is determined to be greater than 0°C ., but less than a predetermined threshold “X”, which may be in the range of $+2$ to $+5^{\circ}\text{C}$. in exemplary embodiments, it may be determined that condensation may be occurring. Where the dewpoint spread is determined to be greater than $X^{\circ}\text{C}$., it may be determined with a high degree of confidence that condensation will not occur. Modified operations may be undertaken where it is determined that condensation is occurring or may be occurring according to the above-described criteria, but not when condensation is not occurring according to the above-described criteria. Determining which of the interior surfaces is coldest, and thus potentially having condensation, may be difficult to determine. Thus, modified operations may be conservatively undertaken for both where condensation is occurring and where condensation may be occurring according to the above-described criteria.

(11) In exemplary embodiments, without limitation, where the dewpoint spread is greater than or equal to $X^{\circ}\text{C}$., which is variable but in exemplary embodiments may be 2°C ., between 2°C .- 5°

C., 4° C., or 5° C. for example, without limitation, a determination may be made that no condensation is likely. Where the dewpoint spread is greater than Y° C., which is variable but in exemplary embodiments may be 0° C., between 0° C.-2° C., or 2° C. for example, without limitation, and less than X° C., a determination may be made that condensation may or may not be present. Where the dewpoint spread is less than or equal to Y° C., a determination may be made that condensation is likely or definitely present within the electronic display assembly. In exemplary embodiments, the controller may be configured to operate the assembly normally (e.g., without any modified operations to minimize, reduce, control, and/or eliminate condensation formation within the electronic display assembly) where the determination is made that no condensation is likely, and may initiate certain modified operations where the determination is made that condensation may or may not be present and/or that condensation is likely present. Such modified operations may be configured to minimize, reduce, control, and/or eliminate condensation formation within the electronic display assembly.

(12) Alternatively, or additionally, a single dewpoint spread threshold may be utilized such that the units are programmed to operate in either a normal operating mode or a condensation mitigation mode. For example, without limitation, where the dewpoint spread is greater than “X” the unit may operate normally, but where the dewpoint spread is less than “X” the unit may operate in condensation mitigation mode. When equal to “X” the unit may be configured to either operate in the normal mode or the condensation mitigation mode. In exemplary embodiments, the threshold for activating condensation mitigation mode may be different from deactivating condensation mitigation mode. For example, the threshold for transitioning from normal operating mode to condensation mitigation mode may be X-A (e.g., 3° C. by way of non-limiting example) while the threshold for transitioning from condensation mitigation mode to normal operating mode may be X+B (e.g., 6° C. by way of non-limiting example). An electronic notification may be generated and transmitted where the dewpoint spread reaches a predetermined threshold below “X” (e.g., X-C, such as 0° C. by way of non-limiting example).

(13) In other exemplary embodiments, where a determination is made that condensation may or may not be present and/or that condensation is likely present (e.g., dewpoint spread less than X° C., or less than or equal to X° C.), a determination may be made as to whether certain safety thresholds are met and/or exceeded. If the safety thresholds are met and/or exceeded, the electronic display assembly may be configured to operate normally (e.g., without modified, condensation control operations). In this way, the operational safety of the display assembly and/or its components may be prioritized. If the safety thresholds are not met and/or exceeded, the electronic display assembly may operate under the modified operating parameters configured to minimize, control, reduce, and/or eliminate condensation within the electronic display assembly.

(14) In yet other exemplary embodiments, modified operations may be undertaken based on a dewpoint threshold and buffer, with or without the secondary check for safety thresholds. For example, without limitation, a threshold dewpoint spread of A° C. and a buffer of B° C. may be set. Where the dewpoint spread exceeds A° C. by more than B° C., modified operations may be undertaken. In exemplary embodiments, a check to ensure that the safety thresholds are not met or exceeded may be first undertaken. A° C. and/or B° C. may each be independent variables and may be, for example, without limitations, each 2° C., A=4° C. and B=2° C., A=5° C. and B=1-4° C., combinations thereof, or the like.

(15) The safety thresholds may comprise any temperature or other condition of any components. For example, without limitation, the safety thresholds may be designed to prevent overheating of sensitive, critical, and/or expensive electronic components. Modified operations, as explained more fully below, may reduce or eliminate ambient air introduction and/or promote heat generation within the unit. The normal operations may permit the ingestion of increased or unlimited amounts of ambient air, which may be prioritized over condensation where the safety thresholds are met and/or exceeded. Where such safety thresholds are not met and/or exceeded, condensation control

may be prioritized.

(16) Alternatively, or additionally, the safety thresholds may not be required.

(17) Modified operations may include, for example, without limitation, increasing heat in the electronic display assembly, such as, but not limited to, by changing, such as by reducing, fan speed, operating time, combinations thereof, or the like and/or increasing power to lighting elements, restricting the ability to turn off or reduce power to the lighting elements, combination thereof, or the like. Modified operations may reduce, prevent, control, and/or eliminate the formation of condensation within the display assembly by decreasing the ingestion of outside (ambient) air, which may be relatively cold (e.g., turning ambient air fans off and/or minimizing their operation) and/or increasing power to lighting elements (e.g., turning on, increasing power, preventing dimming).

(18) For non-emissive displays, such as LCDs, overall image luminance may be kept low as desired (e.g., no noticeable increase in visible image luminance) by turning backlight power up while concurrently turning image gray scale down, for example, without limitation. For non-emissive or emissive displays, power consumption may alternatively, or additionally, be increased by turning drive current to maximum while turning pulse width modulation (PWM) control to minimum levels required to maintain desired perceptive brightness, for example, without limitation. In exemplary embodiments, modified operations may be performed once internal and/or ambient temperatures, dewpoint, relative humidity, and/or dewpoint spread reach a certain threshold, range, combination thereof, or the like and/or where certain safety thresholds are not yet met and/or exceeded.

(19) Such condensation controls may be particularly useful where the electronic display assembly is powered down or otherwise placed in a reduced power mode and/or when ambient temperatures drop, one or both of which may occur during nighttime hours, winter hours, and/or under power efficiency efforts to name a few examples. In exemplary embodiments, without limitation, the systems and methods shown and/or described herein may accomplish condensation control without the need for a separate and/or dedicated heater, thereby reducing power consumption and/or noise.

(20) The various criteria described herein, including, but not limited to, the dewpoint spread ranges and safety thresholds, are merely exemplary and are not intended to be limiting. Other criteria and/or thresholds may be utilized. For example, without limitation, other data points, dewpoint spread criteria, operation modifications, temperatures, thresholds, safety thresholds, and/or ranges, may be utilized.

(21) Alternatively, or additionally, it may be desirable to operate air circulation devices within the display assembly to control temperatures within the display assembly and/or provide condensation control. In exemplary embodiments, the air circulation devices may comprise fan units, each of which may comprise one or more fans, and may be associated with one or more sensors, such as temperature sensors. Zones may be virtually defined within the display assemblies, each of which may include one or more of the air circulation devices and one or more of the associated sensors.

(22) Operational ranges for the air circulation devices may be established. Such operational ranges may be programmed at, or stored at, the controller(s). Such operational ranges may be stored in association with one or more of the air circulation devices. Desired operating ranges may be established for the sensors. Such desired operating ranges may be programmed at, or stored at, the controller(s). Such desired operating ranges may be stored in association with one or more of the sensors.

(23) Operational ranges for the air circulation devices and/or desired operating ranges for the sensors may be specific to the date, time, ambient conditions, combinations thereof, or the like. Operational ranges for the air circulation devices and/or desired operating ranges for the sensors may be specific to the zone, air circulation device, and/or sensor or for the whole display assembly.

(24) Readings from the sensors may be taken periodically, continuously, sporadically, or the like. Operation of some or all of the air circulation devices may be controlled by the highest sensor

reading relative to the associated desired operating range. Such control may be performed on a zone-by-zone basis or for the entire display assembly.

(25) Where a maximum operating temperature is reached or exceeded at one or more of the sensors, speed or other operating conditions (e.g., number of active fans, volumetric flow rate, power supplied, etc.) of the air circulation devices may be increased, such as by the controller(s), until a maximum operational level is reached. If the maximum operating temperature is reached or exceeded at one or more of the sensors, power to the backlight may be reduced until temperatures fall below the maximum operating temperatures. Such reduction may be made in an inversely proportional fashion to how far the temperature has exceeded the maximum operating temperature.

(26) Sensor readings may be continuously or periodically retaken and operations adjusted accordingly.

(27) One or more gaskets or other sealing devices for separating an open loop and/or ambient environment from a closed loop or at least partially sealed area may be provided between a portion of the housing for the unit (e.g., a framework or chassis) and a cover glass so as to reduce or prevent thermal transfer through the portion of the housing for the unit, which may comprise metal.

(28) Further features and advantages of the systems and methods disclosed herein, as well as the structure and operation of various aspects of the present disclosure, are described in detail below with reference to the accompanying figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In addition to the features mentioned above, other aspects of the present invention will be readily apparent from the following descriptions of the drawings and exemplary embodiments, wherein like reference numerals across the several views refer to identical or equivalent features, and wherein:

(2) FIG. 1 is a perspective view of an exemplary display assembly in accordance with the present invention and indicating section lines A-A and B-B;

(3) FIG. 2A is a detailed side sectional view of the display assembly of FIG. 1 taken along section line A-A and indicating detail A;

(4) FIG. 2B is a detailed perspective view of detail A of FIG. 2A;

(5) FIG. 2C is a detailed side sectional view of another exemplary embodiment of the display assembly of FIG. 1 taken along section line A-A;

(6) FIG. 3 is a simplified top sectional view of the display assembly of FIG. 1 taken along section line B-B;

(7) FIG. 4 is a flow chart with exemplary logic for operating the electronic display assembly of FIGS. 1-3 in accordance with the present invention;

(8) FIG. 4B is a flow chart with other exemplary logic for operating the electronic display assembly of FIGS. 1-3 in accordance with the present invention;

(9) FIG. 5 is a flow chart with other exemplary logic for operating the electronic display assembly of FIGS. 1-3 in accordance with the present invention;

(10) FIG. 6 is a plan view of an exemplary operations programming interface for the electronic display assemblies of FIGS. 1-5;

(11) FIG. 6B is a plan view of another exemplary operations programming interface for the electronic display assemblies of FIGS. 1-5;

(12) FIG. 7A is a flow chart with other exemplary logic for operating the electronic display assembly of FIGS. 1-6 in accordance with the present invention;

(13) FIG. 7B is a flow chart with other exemplary logic for operating the electronic display assembly of FIGS. 1-7A in accordance with the present invention;

- (14) FIG. 7C is a flow chart with other exemplary logic for operating the electronic display assembly of FIGS. 1-6 in accordance with the present invention; and
- (15) FIG. 8 is a flow chat with exemplary logic for operating the electronic display assembly of FIGS. 1-7B in accordance with the present invention.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENT(S)

(16) Various embodiments of the present invention will now be described in detail with reference to the accompanying drawings. In the following description, specific details such as detailed configurations and components are merely provided to assist the overall understanding of these embodiments of the present invention. Therefore, it should be apparent to those skilled in the art that various changes and modifications of the embodiments described herein can be made without departing from the scope and spirit of the present invention. In addition, descriptions of well-known functions and constructions are omitted for clarity and conciseness.

(17) Embodiments of the invention are described herein with reference to illustrations of idealized embodiments (and intermediate structures) of the invention. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments of the invention should not be construed as limited to the particular shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

(18) FIG. 1 through FIG. 3 illustrate an exemplary electronic display assembly **10**. The assembly **10** may comprise a housing **18**. The assembly **10** may comprise a cover panel **12**. The cover panel **12** may comprise a single layer or multiple layers **12A**, **12B** bonded together, such as by way of one or more optical adhesives. The cover panel **12** may be located forward of, and may be spaced apart from, an electronic display layer **14**. The electronic display layer **14** may comprise a layer of liquid crystals (e.g., an LCD), a plasma display, OLED display, LED display, combinations thereof, or the like. The cover panel **12** may form a forward portion of the housing **18**. The housing **18** may substantially enclose the electronic display layer **14**.

(19) The cover panel **12** may be transparent or translucent such that images displayed at the electronic display layer **14** are visible to an intended viewer through the cover panel **12**. The cover panel **12** may be configured to protect the electronic display layer **14** and/or other components of the electronic display assembly **10**. The cover panel **12** may, alternatively, or additionally, be configured to enhance optics of the images displayed at the electronic display layer **14**. The cover panel **12** and/or electronic display layer **14** may comprise one or more polarizers, anti-reflective films, surface treatments, combinations thereof, or the like. A front air gap **13** may be located rearward of the cover panel **12** and forward of the electronic display layer **14**. The front air gap **13** may form part of a closed loop airflow pathway for circulating gas **58**.

(20) An illumination device **16** may be provided adjacent to at least a portion of the electronic display layer **14**. The illumination device **16** may comprise a number of lighting elements **38**. The lighting elements **38** may comprise light emitting diodes (LEDs), though other kinds or types of lighting elements **38** may be utilized. The illumination device **16** may be configured to provide illumination to the electronic display layer **14** when powered. For example, without limitation, the illumination device **16** may be configured to provide direct backlight for the electronic display layer **14** and may be positioned rearward of the electronic display layer **14**. Alternatively, or additionally, the illumination device **16** may be configured to provide edge lighting for the electronic display layer **14** and may be positioned around some or all of a perimeter of the electronic display layer **14**, one or more light guides, reflective elements, combinations thereof, or the like. In exemplary embodiments, the illumination device **16** may comprise a number of the lighting elements **38** provided on one or more tiles, mounted to a substrate (e.g., printed circuit board), combinations thereof, or the like. Any number, arrangement, and/or type of the lighting elements **38** may be used.

(21) In other exemplary embodiments, the electronic display layer **14** may be an emissive display

and/or may be configured to illuminate without the need for a separate and/or dedicated illumination device **16**. Examples of such embodiments include, without limitation, OLED displays, plasma displays, LED displays, combinations thereof, or the like.

(22) The assembly **10** may comprise one or more open loop heat exchangers (hereinafter also “OL HX”) **20**. The OL HX **20** may be configured to accommodate ambient air. The OL HX **20** may be provided rearward of the illumination device **16**. In exemplary embodiments, the OL HX **20** may extend along some or all of the illumination device **16** so as to absorb some or all of the heat generated by the illumination device **16** when in use. The OL HX **20** may extend directly along the illumination device **16** or may be spaced apart therefrom. For example, without limitation, one or more thermally conductive layers, air gaps, and/or spacers may be positioned between the illumination structure and the OL HX **20**.

(23) The OL HX **20** may comprise one or more layers. In exemplary embodiments, some or all of the layers of the OL HX **20** may comprise a corrugated structure **26**. The corrugated structure **26** may comprise a zigzag pattern which extends between two or more panels or layers of the OL HX **20**, thereby forming a number of channels or pathways within the OL HX **20**. Alternatively, or additionally, the OL HX **20** may comprise a number of tubes (e.g., square, rectangular, round, combinations thereof, or the like) defining passageways or channels for ambient air.

(24) In exemplary embodiments, the OL HX **20** may be in fluid communication with one or more intakes and exhausts provided in the housing **18**. Such intakes and exhausts may comprise one or more apertures in the housing **18** which permit the intake and exhaust, respectively, of ambient air from the assembly **10**.

(25) One or more air circulation devices **48**, such as, but not limited to, fans may be provided within or otherwise in fluid communication with the OL HX **20** and/or other portions of one or more open loop airflow pathways within the electronic display assembly **10** to cause the ingestion of ambient air into the assembly **10**, flow through the one or more open loop airflow pathways, and exhaustion from the assembly **10** when operated. Such air circulation devices **48** may be in electronic communication with one or more controller(s) **52**. The air circulation devices **48** may comprise, for example, without limitation, axial fans, centrifugal fans, combinations thereof, or the like. Any number and/or type of air circulation devices **48** at any one or number of locations within the display assembly **10** may be utilized.

(26) The assembly **10** may comprise one or more open loop/closed loop heat exchangers (hereinafter also “OL/CL HX”) **22**. The OL/CL HX **22** may be provided rearward of the illumination device **16**. The OL/CL HX **22** may comprise multiple layers. In exemplary embodiments, each of the layers may be configured to accommodate one of: only ambient air as part of one or more of the open loop airflow pathways in the assembly **10**, or circulating gas as part of one or more closed loop airflow pathways in the assembly **10**. The layers may be arranged, for example, without limitation, vertically or horizontally adjacent one another. For example, without limitation, the layers may alternate between being configured to accommodate ambient air and circulating gas. In exemplary embodiments, a first portion of the layers may form part of the same or a different open loop airflow pathway as the OL HX **20** and a second portion of the layers may form part of the same or a different closed loop airflow pathway as the front air gap **13**. The OL/CL HX **22** may be in fluid communication with the same or different intake(s) and exhaust(s) as the OL HX **20**. The OL/CL HX **22** may be in fluid communication with the front air gap **13**, though such is not required.

(27) In exemplary embodiments, the electronic display assembly **10** may comprise multiple electronic display layers **14**. In such embodiments, the electronic display assembly **10** may comprise multiple cover panels **12**, illumination structures **16**, OL HX **20**, OL/CL HX **22**, combinations thereof, or the like. However, at least the OL/CL HX **22** may be common to multiple electronic display layers **14** in some embodiments. For example, without limitation, the electronic display assembly **10** may comprise a first and second electronic display layer **14** provided in a

back-to-back arrangement with front air gaps **13** fluidly connected to a common OL/CL HX **22** but separate OL HXs **20** for each electronic display layer **14**. Such electronic display layers **14** may be provided in the same or different housings **18**.

(28) In exemplary embodiments, the electronic display layer **14**, the illumination device **16**, and OL HX **20** may be provided within an access assembly **44** along with one or more components, such as, but not limited to, electronic circuits, air circulation devices **48**, sensors **54**, controller(s) **52**, power supplies, wiring, processors, video players, cameras, microphones, combinations thereof, and the like. The access assembly **44** may comprise a housing, framework, or one or more structural members, which may be attached to the housing **18** by way of one or more movement devices **46**, such as, but not limited to, hinges, springs, levers, pistons, combinations thereof, or the like configured to facilitate movement of the access assembly **44** between an opened position where the access assembly **44** is moved away from the housing **18** and a closed position where the access panel **44** is adjacent the housing **18**. One or more sealing devices **42**, such as, but not limited to, gaskets, may be provided between the housing **18** and the access assembly **44** to partially or completely seal when said access assembly **44** is placed in the closed position.

(29) As illustrated with particular regard to at least FIG. 2C, the sealing devices **42**, such as but not limited to gaskets, may be provided between the cover panel **12** and the housing **18**. For example, without limitation, the sealing devices **42**, such as but not limited to gaskets, may extend at a first, preferably free end, to an interior surface of the cover panel **12** and may extend at a second, preferably fixed end, to a portion of the housing **18**. Stated another way, the cover panel **12** may extend above and below where the sealing device **42** meets the cover panel **12**. This exemplary arrangement, particularly placement of the sealing device **42** against the cover panel **12** instead of the housing **18**, may assist with reducing thermal transfer from external, ambient temperatures into the units **10**. In particular, because metal (of which the housing **18** may be comprised of in whole or in part) has a relatively high level of thermal conductivity where glass and/or polymer (of which the cover panel **12** may be comprised of in whole or in part) has a relatively low level of thermal conductivity. Thus, relatively cool temperatures of ambient air transferred to the housing **18** are less likely to be thermally conducted into relatively warm areas of the closed loop, by way of non-limiting example. This may reduce “cold spots” within the unit **10** where condensation is otherwise likely to form. The cover panel **12**, particularly when comprising glass and/or a polymer, may also provide a smoother surface for sealing, thereby providing a better seal, particularly compared to areas of metal where adhesive, welding, grinding, or the like may be provided.

(30) In exemplary embodiments, such sealing devices **42** may be liquid impermeable, but may be vapor permeable. When moved into the open position, certain components of the electronic display assembly **10**, such as, but not limited to, a rear area of the OL HX **20**, the OL/CL HX **22**, customer equipment, server racks, storage compartments, combinations thereof, or the like may be accessible. Furthermore, the access assembly **44** may be removed for service and/or replacement. Where more than one electronic display layer **14** is utilized, more than one access assembly **44** with the same or similar components may be provided and connected to the housing **18**.

(31) Ambient air **56** may extend through one or more open loop airflow pathways within one or more of the OL HX **20**, the OL/CL HX **22**, combinations thereof, or the like. Such open loop airflow pathway(s) may be partially, mostly, substantially, or entirely sealed to separate ambient air **56** from circulating gas **58** traveling through one or more closed loop airflow pathways within one or more of the front air gap **13**, the OL/CL HX **22**, within the housing **18**, combinations thereof, or the like. In this way, particularly the ambient air **56** may be kept partially, mostly, substantially, or entirely separate from the circulating gas **58**.

(32) One or more air circulation devices **48**, such as, but not limited to, fans, may be provided within or otherwise in fluid communication with the OL/CL HX **22**, the front air gap **13**, and/or other portions of one or more closed loop airflow pathways within the electronic display assembly **10** to cause the flow of circulating gas through some or all of the same when the air circulation

devices **48** are operated. Such air circulation devices **48** may be in electronic communication with the same or different ones of the one or more controller(s) **52**. The air circulation devices **48** may comprise, for example, without limitation, axial fans, centrifugal fans, combinations thereof, or the like. Any number and/or type of air circulation devices **48** at any one or number of locations within the display assembly **10** may be utilized.

(33) Alternatively, or additionally, the display assemblies **10** may comprise heaters, air conditioning units, filters, thermoelectric modules, heat sinks, combinations thereof, or the like.

(34) The electronic display assembly **10** may comprise one or more sensors **54**. The sensors **54** may be provided at one or more locations at the electronic display assembly **10**. Some or all of the sensors **54** may be in electronic communication with one or more of the same or different controller(s) **52**. Such sensors **54** may include, for example, without limitation, location detection devices. Alternatively, or additionally, location data may be pre-programmed or updated manually. The sensor **54** may include temperature sensors, which may be located at intake(s), along the one or more open loop airflow pathways, at the exhaust(s), within the closed loop airflow pathways, at the illumination device **16**, at power supplies (which may be, for example, without limitation, located at and/or along rear surfaces of the illumination device **16**), at FPGA (Field Programmable Gate Array) die, at various framework or other components of the electronic display assembly **10**, at the one or more controller(s) **52**, a FPGA die, processor board, combinations thereof, or the like.

(35) In exemplary embodiments, without limitation, at least one temperature and/or humidity sensor is located at, or proximate, an upper or uppermost surface of the housing **18** portion defining the closed loop airflow pathway, and at least one other temperature and/or humidity sensor is located at, or proximate, a lower or lowermost surface of the closed loop airflow pathway the housing **18** portion defining the closed loop airflow pathway. These may be, by way of non-limiting example, the coldest portions of the unit **10**, and thus the most likely to experience condensation. Using data from these locations may provide particularly conservative measurements for condensation control.

(36) Alternatively, or additionally, temperature and/or humidity sensors may be located at, or proximate, to an intake area, such as to detect the temperature of ingested ambient air.

(37) These exemplary arrangements, and others, may assist with providing accurate information for determining internal unit **10** conditions, such as dewpoint spread for initiating modified operations. However, any number and location of temperature and/or humidity sensors may be utilized.

(38) In exemplary embodiments, without limitation, multiple temperature readings may be taken at various portions of the assembly **10**. Such values may be compared, such as but not limited to with respective offsets, and the lowest (coldest) values may be utilized by the controller(s) **52** to control the assembly **10** for purposes of operating the unit **10** normally or in the condensation mitigation mode. Offsets may be established to adjust for perceived inaccuracies, likely temperatures at other points in the units, localized temperature goals or needs, combinations thereof, or the like.

(39) The sensors **54** may comprise one or more humidity sensors, which may be provided at the one or more open loop airflow pathways, the one or more closed loop pathways, one or more components of the display assembly **10**, combinations thereof, or the like. The various sensors **54** may be configured to report readings data to the controller(s) **52**. The size, shape, and/or location of the sensors **54** shown and/or described are merely exemplary and are not intended to be limiting. Any type, kind, and/or number of sensors **54** may be provided at any number of locations within the display assembly **10** to measure any number or type of data points. The humidity sensors, which may be configured to determine relative humidity, may include temperature sensors and absolute humidity sensors.

(40) While illustrated internal to the display assembly **10**, one or more of the sensors **54** and/or controller(s) **52** may be external to the display assembly **10**. For example, without limitation, one or more of the sensors **54** may be located outside the housing **18**. As another example, without limitation, some or all of the data points may be retrieved over one or more networks, such as the

world wide web, from remote weather stations. The display assembly **10** may comprise one or more network communication devices **62** configured to retrieve such data, which may be periodically or continuously updated. As another example, without limitation, the controller(s) **52** may comprise one or more remote monitoring and/or control systems, such as, but not limited to, computers, smartphones, tablets, servers, combinations thereof, or the like, which may be in electronic communication with one or more controller(s) **52**, processors, combinations thereof, or the like by way of one or more network connectivity devices. Data from both sensors **54** at the display assembly **10** and retrieved from outside sources may be utilized. For example, without limitation, data from outside sources may be retrieved by way of the network communication devices **62**.

(41) In exemplary embodiments, at least the controller **52** and/or network communication devices **62** are provided at the closed loop airflow pathways, though any location may be utilized.

(42) FIG. **4** illustrates a flow chart with exemplary logic for operating the electronic display assembly **10** to control condensation. The relative humidity of air inside the display assembly **10** may be measured. For example, without limitation, the relative humidity may be determined by sampling data from the sensor(s) **54** configured to measure relative humidity at the display assembly **10**. Alternatively, or additionally, the relative humidity may be determined by sampling data from the sensor(s) **54** configured to measure humidity at the display assembly **10** and the sensor(s) **54** may configured to measure temperature at the display assembly **10**, which may be the same or different from one another, and the relative humidity within the display assembly **10** may be calculated or determined from such data. Such calculation or determination may be made at the sensor(s) **54** or the controller(s) **52**, such as based on a table or other data sources. Such temperature and/or humidity readings may be determined from sensor(s) **54** located at the one or more open loop airflow pathways of the ambient air, the one or more closed loop airflow pathways of the circulating gas, combinations thereof, or the like. Alternatively, or additionally, the temperature, humidity, and/or relative humidity may be retrieved from one or more network sources based on reported and/or measured conditions proximate a location of the electronic display assembly **10**.

(43) The dewpoint may be calculated from the relative humidity, humidity, and/or certain temperature data. The dewpoint may be calculated at the controller(s) **52** and/or sensor(s) **54**, such as, for example, without limitation, by using formulas available at:

<http://bmcnoldy.rsmas.miami.edu/Humidity.html> and/or <https://bmcnoldy.rsmas.miami.edu/mia/>. Any formula or algorithm for calculating dewpoint, known or yet to be developed, may be utilized. Alternatively, or additionally, the dewpoint may be retrieved from one or more network sources based on reported and/or measured conditions proximate the location of the electronic display assembly **10**. The temperature used to calculate the dewpoint may be determined from sensor(s) **54** located at the one or more open loop airflow pathways of the ambient air, the one or more closed loop airflow pathways of the circulating gas, one or more network sources based on reported, measured conditions proximate a location of the electronic display assembly **10**, combinations thereof, or the like.

(44) The dewpoint spread may be calculated, such as by way of the controller(s) **52**. The dewpoint spread may be calculated between the certain temperature data and the dewpoint. The temperature may be determined from sensor(s) **54** located at the one or more open loop airflow pathways of the ambient air, the one or more closed loop airflow pathways of the circulating gas, one or more network sources based on reported, measured conditions proximate a location of the electronic display assembly **10**, combinations thereof, or the like. In exemplary embodiments, without limitation, the temperature may be determined as the lessor of sensor(s) **54** readings from within the electronic display assembly **10** or retrieved temperature conditions as retrieved from one or more network sources based on reported and/or measured conditions proximate the location of the electronic display assembly **10**. Data from network sources shown and/or described herein may be

retrieved, for example, without limitation, by way of the one or more network communication devices **62**.

(45) If the dewpoint spread is greater than or equal to $X^{\circ}\text{C.}$, which is variable but may be 2°C. , between 2°C. and 5°C. , 4°C. , or 5°C. for example, without limitation, a determination may be made, such as at the controller(s) **52**, that no condensation is likely present. In such cases, the controller(s) **52** may be configured to operate the electronic display assembly **10** normally, such as under default operating parameters. If the dewpoint spread is greater than $Y^{\circ}\text{C.}$, which is variable but may be set of 0°C. , between 0°C. and 2°C. , or 2°C. in exemplary embodiments, and less than $X^{\circ}\text{C.}$, a determination may be made that condensation may or may not be present. In such cases, the controller(s) **52** may be configured to operate the electronic display assembly **10** in a first modified operating mode. If the dewpoint spread is less than or equal to $Y^{\circ}\text{C.}$, a determination may be made that condensation is likely present. In such cases, the controller(s) **52** may be configured to operate the electronic display assembly **10** in a second modified operating mode. The second modified operating mode may be the same or different from the first modified operating mode.

(46) Other criteria, ranges, and/or thresholds may be utilized. For example, without limitation, other dewpoint spread criteria, ranges, and/or thresholds may be utilized. For example, without limitation, where a dewpoint spread equal to or less than $X^{\circ}\text{C.}$ is determined, the controller(s) **52** may be configured to command modified operations. As another example, without limitation, a half, quarter, or full degree margin of error may be utilized such that dewpoint spread equal to or less than $X+1^{\circ}\text{C.}$, by way of non-limiting example, may result in the controller(s) **52** commanding modified operations. These are just examples and are not intended to be limiting. Any number of thresholds and/or ranges and modified operating modes may be utilized.

(47) The modified operating mode(s) (including, but not limited to, the first and second modified operating modes) may include commanding certain actions, such as, but not limited to, by way of the controller(s) **52**, configured to raise the temperature of the electronic display assembly **10**, thereby reducing the likelihood of condensation, and/or drive out moisture in the electronic display assembly **10**. In exemplary embodiments, moisture may be driven out by increasing the temperature which may cause the air within the assembly, such as, but not limited to, the circulating gas **58** in the one or more closed loop airflow pathways, to expand. Because the closed loop airflow pathways are otherwise fully or partially sealed, this may result in driving out at least a portion of the circulating gas **58** from the one or more closed loop pathways, which may bring vaporized moisture with it. The air may permeate through one or more gaskets, which may be liquid tight but not necessarily vapor tight. In exemplary embodiments, the increased heat and/or airflow, such as from the one or more air circulation devices **48**, may cause liquid moisture to vaporize or otherwise be gathered into the circulating gas which is then driven out of the display assembly **10**.

(48) The modified operating mode(s) may comprise commands to increase illumination at the illumination device **16**, such as, but not limited to, driving the lighting elements **38** at an increased power level, reduce local dimming, and/or reduce dynamic dimming of the illumination device **16**. This may be accomplished with or without altering operation of the electronic display layer **14**. For example, without limitation, the electronic display layer **14** may be commanded to show a blank black screen despite increased illumination to increase heat without causing light pollution. As another example, without limitation, certain parameters of the electronic display layer **14**, such as, but not limited to, grayscale may be altered to maintain essentially the same visible image characteristics while increasing the illumination. In these ways, the image displayed may appear unaltered to viewers, may conform to customer requirements, and/or prevent or reduce light pollution for example, without limitation. Alternatively, or additionally, limiting light leakage may maximize heat retention within the assembly **10**.

(49) The modified operating mode(s) may, alternatively, or additionally, comprise commands to

reduce speed of and/or cease or otherwise minimize operation of air circulation devices **48** associated with the open loop airflow pathway(s). This may reduce the amount of relatively cool ambient air ingested into the assembly **10**, which may cause temperatures to rise, or at least not lower as quickly. Operations of the air circulation devices **48** associated with the closed loop airflow pathway(s) may be modified as well. For example, without limitation, commands to increase the speed or and/or operation of such air circulation devices **48** may be increased, which may cause the circulating gas **58** to pick up condensation moisture.

(50) In exemplary embodiments, once adequate temperatures, relative humidity, dewpoint, and/or dewpoint spread is reached, the modified operating mode(s) may be ceased and/or normal operations may be resumed. In exemplary embodiments, the commands shown and/or described herein may be carried out by the controller(s) **52**.

(51) Normal operating mode may be default mode or otherwise preprogrammed operating parameter, conditions, and/or logic. Such normal operating mode may permit relatively higher or unlimited air circulation device **48** speeds, run times, combinations thereof, or the like, particularly, but not limited to, for those air circulation devices **48** associated with the open loop airflow pathways.

(52) Stated another way, where the dewpoint spread is less than X and/or Y (or another threshold), a determination may be made that the conditions are such that condensation is more likely to form. The modified operations, either of the first and/or second variety, may be automatically initiated by software routine which have the goal of increasing the dewpoint spread and/or otherwise mitigate the likelihood of condensation forming within the assemblies **10**. Such modified operations may include adjusting the speed and/or thresholds at which air circulation devices **48** operate, adjusting illumination device **16** operations, combinations thereof, or the like. This may include increasing or decreasing fan speed. Such alterations may be made to acceptable fan speed ranges (e.g., 0-75% speed, 25-95% fan speed, etc.), fan speed limits (<25%, over 50 rpm, etc.), and/or operational speed setpoints (e.g., zero speed, 50% speed, 100% speed, etc.).

(53) As illustrated with particular regard to at least FIG. **4B**, alternatively or additionally, a single dewpoint spread threshold may be used such that the units **10** are configured to operate, such as by way of programmable control of the controller(s) **52**, in either a normal operating mode or a condensation mitigation mode. For example, without limitation, where the dewpoint spread is determined, such as by the controller(s) **52**, to be greater than “X” the controller(s) **52** may configure the unit **10** to operate normally, but where the dewpoint spread is determined to be less than “X” the controller(s) **52** may configure the unit **10** to operate in the condensation mitigation mode. When equal to “X”, the controller(s) **52** may configure the unit **10** to operate may be configured to either operate in the normal mode or the condensation mitigation mode. The controller(s) **52** may be configured to set the dewpoint spread to a threshold, such as between 1° C. and 10° C. The controller(s) **52** may configure cause the unit **10** to remain in the condensation mitigation mode unless and until the dewpoint spread increases above X by at least a predetermined margin, such as between 1° C. and 5° C. Alternatively, or additionally, the threshold for activating condensation mitigation mode may be different from deactivating condensation mitigation mode. For example, the threshold for transitioning from normal operating mode to condensation mitigation mode may be X-A (e.g., 3° C.) while the threshold for transitioning from condensation mitigation mode to normal operating mode may be X+B (e.g., 6° C.). An electronic notification may be generated and transmitted where the dewpoint spread reaches a predetermined threshold below “X” (e.g., X-C, such as 0° C.). This may reduce or prevent rapid transitions between condensation mitigation mode and normal operating mode.

(54) FIG. **5** illustrates a flow chart with other exemplary logic for operating the electronic display assembly **10** to minimize, reduce, control and/or eliminate condensation. The steps shown and/or described with respect to FIG. **5** may be the same or similar to those shown and/or described with respect to FIG. **4** except as otherwise specified herein. For example, without limitation, if a determination is made that condensation may or may not be present or that condensation is likely

present, a determination may be made, such as at the controller(s) **52**, whether certain safety thresholds are met and/or exceeded. If such safety thresholds are met and/or exceeded, the controller(s) **52** may command the electronic display assembly **10** to operate normally (i.e., without modified operations configured to minimize, reduce, control, and/or eliminate condensation). If some or all of the safety thresholds are not met and/or exceeded, the controller(s) **52** may command the electronic display assembly **10** to initiate modified operations, which may be the same, or different from the modified operations shown and/or described with respect to FIG. **4**.

(55) FIG. **6** illustrates an exemplary interface **64**. The interface **64** may be provided at the controller(s) **52** and/or one or more remote electronic devices. The interface **64** may be configured to accept certain user input regarding safety thresholds, modified operations, normal operations, and other operations of the assembly **10**. For example, without limitation, such criteria may be selected at one or more remote electronic devices, such as, but not limited to, personal computers, server computers, smartphones, tablets, combinations thereof, or the like and transmitted to the assemblies **10** by way of the network communication devices **62**.

(56) In exemplary embodiments, the controller(s) **52** may be configured to first temporarily initiate normal operations upon determination that the safety thresholds are met and/or exceeded. This may cause air circulation devices **48**, such as those associated with the open loop airflow pathways, to partially or fully operate at higher levels (e.g., speed, runtimes, etc.), ingesting relatively more ambient air in attempt to cool the assembly **10** for a period of time. Normal operations may be resumed for a period of time, such as, but not limited to, 120 seconds, though the amount of time may be variable and may be programmed and/or altered, such as at value **63**. If after the period of time safety thresholds are no longer met and/or exceeded, modified operations may be resumed. If the safety thresholds are still met and/or exceeded after the period of time, then the assemblies **10** may be configured to continue normal operations for at least a period of time, which may be fixed or indefinite. The period of time may be programmed at the time parameter of the interface **64**, for example, without limitation. The safety thresholds may be set, for example, without limitation, at the one or more safety parameters **68** of the interface **64**. In exemplary embodiments, only a single temporary initiation of normal operations may be permitted before a longer term or fixed return to normal operations is commanded, such as by the controller(s) **52**.

(57) The source for dewpoint calculations may be set at a dewpoint source parameter **70** of the interface **64**. For example, without limitation, the source may be selected between an internal relative humidity sensor, remote sources, combinations thereof, or the like.

(58) Any of the variables, parameters, conditions, combinations thereof, or the like may be pre-programmed and/or programmed at the same or different interfaces **64**, such as, but not limited to, at the controller(s) **52** and/or by remote devices and the network communication devices **62**.

(59) FIG. **6B** illustrates another exemplary embodiment of the interface **64**. The interface may include various offset values **71**. Offset values **71** may be values added to measured or determined values, such as at the controller(s) **52**, for comparison against a respective threshold. These may be set to reflect errors in measurement, differences from perceived values elsewhere, combinations thereof, or the like. A delay value **73** may be set which delays, such as by way of the controller(s) **52**, implementation of condensation mitigation mode, such as for a specified number of seconds, minutes, combinations thereof, or the like. This may allow the assembly **10** some time to adjust to relatively minor variations without changing operating modes, eliminate erroneous readings, combinations thereof, or the like. A hysteresis value **75** may be established which acts as a margin which the dewpoint spread must clear before the controller(s) **52** switch operating modes. This may prevent rapid or frequent changing between operating modes. A dewpoint spread minimum value **75** may be abolished. This may be dewpoint spread value at which the controller(s) **52** is configured to automatically generate and/or transmit an electronic notification, which may indicate conditions likely to result in condensation. All of the values shown and/or described herein may be set by software program, may be variable, and/or may be locally and/or remotely updated from

time, such as by receipt of one or more authenticated commands.

(60) As further illustrated in FIG. 7A, in exemplary embodiments, the controller(s) 52 may be configured to initiate modified or normal operations based upon a dewpoint spread threshold parameter 74 and/or a buffer 72, one or both of which may be programmable at the interface 64. The dewpoint spread threshold parameter 74 and/or buffer 72 may be utilized to determine a dewpoint spread, such as is shown and/or described herein. The controller(s) 52 may be configured to command the assembly 10 to utilize modified operations where the determined dewpoint spread meets the dewpoint spread threshold parameter 74 and/or exceeds the dewpoint spread threshold parameter 74 by at least, or more than, the buffer 72.

(61) The controller(s) 52 may be configured to command the assembly 10 to continue utilizing modified operations until such a time as the determined dewpoint spread falls below the dewpoint spread threshold parameter 74, such as by more than the buffer 72. The controller(s) 52 may be configured to command the assembly 10 to utilize normal operations where the determined dewpoint spread does not meet the dewpoint spread threshold parameter 74 and/or is below the dewpoint spread threshold parameter 74 by at least the buffer 72. The controller(s) 52 may be configured to command the assembly 10 to continue utilizing normal operations until such a time as the determined dewpoint spread exceeds the dewpoint spread threshold parameter 74 by at least the buffer 72. In this way, a programmable buffer may be provided in either direction of temperature change against the programmable threshold. However, the safety thresholds may still be considered and prioritized such that the assembly 10 defaults to normal operations where one or more of the safety thresholds are met and/or exceeded.

(62) Alternatively, or additionally, the dewpoint spread set buffer 72 may comprise the dewpoint spread value at which the controller(s) 52 will cause the assembly 10 to move into condensation mitigation mode. Stated another way, the dewpoint spread set buffer 72 may be the value J, which when dropped below may initiate condensation control measures. The dewpoint spread clear buffer 74 may comprise the dewpoint spread value at which the controller(s) 52 will cause the assembly 10 to return to normal operating mode. Stated another way, the dewpoint spread clear buffer 74 may be the value K, which when raised above may return the unit 10 to normal operations. In this fashion, the value which activates and deactivates condensation mitigation mode may be different.

(63) As further illustrated in FIG. 7B, in exemplary embodiments, an attempt to bring the assembly 10 back below or otherwise within the safety thresholds may be provided, such as by way of the programmable time parameter. The controller(s) 52 may be configured to temporarily operate the assembly 10 in normal operations for the time parameter. If, after the time parameter is reached, and one or more of the safety thresholds are still exceeded or readings are otherwise outside the safety thresholds, then the controller(s) 52 may move the assembly 10 to normal operations until such a time as the threshold parameters 74 are exceeded by at least the buffer 72 and readings are within the safety thresholds. If, after the time parameter is reached, readings are within the safety thresholds, then the controller(s) 52 may resume modified operations until such a time as the threshold parameters 74 are exceeded by at least the buffer 72 or readings are outside the safety thresholds. This may give the assembly 10 a chance to recover from slipped operations and/or essentially disregard a potentially erroneous and/or outlier reading. This attempt may be only completed one time before commanding normal operations. For example, only a single loop of the command logic may be permitted.

(64) The safety thresholds may include, for example, without limitation, temperatures above or below certain thresholds, within certain ranges, combinations thereof, or the like, such as measured by at least certain of the sensors 54. In exemplary embodiments, without limitation, the safety thresholds may include some or all of the following: temperature as measured by one or more sensors 54 within or adjacent to the closed loop airflow pathways being below 30° C., temperature as measured by one or more sensors 54 at or in proximity to the illumination device 16 below 40° C., PS (Power Supply) Max is below 50° C., FPGA is below 70° C., controller(s) 52 is/are below

50° C., sensor(s) 52 measuring humidity is/are below 35° C., combinations thereof, or the like. Other criteria and/or thresholds may be utilized, such as, but not limited to, other temperatures from other locations and/or at different thresholds. The safety thresholds, for example, without limitation, may be used to more accurately determine internal temperature of the display assembly 10 and thus may provide a more accurate determination of the likelihood that condensation is present and thus whether modified operations should be undertaken. The safety thresholds in exemplary embodiments may be variable and programmable, such as, but not limited to, at the one or more safety parameters 68 of the interface 64. The safety parameters 68, dewpoint source parameter 70, dewpoint spread threshold parameter 74, buffer 72, and/or time parameter shown in FIG. 6 represent exemplary operating parameters and are provided for example, without limitation. (65) In exemplary embodiments, modified operations may be provided regularly, such as during transitions from nighttime to daylight hours, following the end of power efficiency modes, during cold temperatures, combinations thereof, or the like. In other exemplary embodiments, modified operations may be dependent on ambient conditions and/or readings from the sensors 54. For example, without limitation, modified operating operations configured to increase the internal temperature of the electronic display assembly 10, such as, but not limited to, increasing power to illumination device 16, reducing speed of air circulation devices 48, during or following relatively cool nighttime hours and/or days or times with relatively cool ambient temperatures, and modified operations configured to drive out moisture, such as, but not limited to, increased speed of air circulation devices 48, during relatively warm daytime hours and/or days or times with relatively warm ambient temperatures.

(66) Modified operations may include changing operational speed of air circulation devices 48 of the unit 10. In exemplary embodiments, without limitation, modified operations may include decreasing operational speed of air circulation devices 48 associated with open loop airflow pathways and/or ambient air, such as down to 20% (of maximum normal operation speed) or zero, though any speed, speed reduction, speed range, or the like may be utilized. In this fashion, flow of typically relatively cooler ambient air may be decreased, which tends to warm the units 10. Alternatively, or additionally, modified operations may include increasing operational speed of air circulation devices 48 associated with closed loop airflow pathways or areas and/or circulating gas, such as up to 80% or 100% (of maximum normal operation speed), though any speed, speed reduction, speed range, or the like may be utilized. This may assist with driving moisture out of closed areas, such as any closed loop airflow pathways, sealed, or semi-sealed areas, homogenizing internal air temperatures, and/or promoting efficient thermal transfer, by way of non-limiting example.

(67) In other exemplary embodiments, the temperature, humidity, relative humidity, dew point spread, or other data may be derived from predicted weather data, such as based on historical patterns, from internet-based sources, combinations thereof, or the like. In such cases, certain modified operating modes may be scheduled and/or initiated as preventative measures based on predicted data.

(68) The various measures shown and/or described herein, including, but not limited to, humidity measures, temperature measures, combinations thereof, or the like, may be determined by way of multiple measurements from the same or different sensors 54, internet-based sources or other remote measures or user input, combinations thereof, or the like, and the utilized measures may be an average, highest, lowest, mean, median value, combinations thereof, or the like.

(69) In exemplary embodiments, some or all air circulation devices 48 may be kept at a minimum speed, such as regardless of normal or modified operations, to provide relatively uniform temperature within the electronic display assembly 10, consistent readings at sensors 52, combinations thereof, or the like. In exemplary embodiments, without limitation, this minimum speed may be 12% of maximum possible normal operating speed.

(70) As illustrated with particular regard to at least FIG. 7C, the safety parameters analysis is

optional. This applies to all embodiments shown and/or described herein.

(71) FIG. 8 provides other exemplary logic for operating the display assembly 10. It may be desirable to operate the air circulating devices 48 within the display assembly 10 to control temperatures within the display assembly 10 and/or provide condensation control. In exemplary embodiments, each of the air circulating devices 48 may comprise one or more fans, such as provided in banks or sets. Each of the air circulating devices 48 may be associated with one or more of the sensors 54. Each of the sensors 54 may be configured to measure temperature of proximate air.

(72) Zones may be virtually defined within the display assemblies 10, such as at the controller(s) 52. Each zone may be associated with one or more of the air circulating devices 48 and/or one or more of the sensors 54. For example, without limitation, one zone may comprise front air gap 13, another zone may include the OL HX 20, another zone may include the OL/CL HX 22, combinations thereof, or the like. Within the front air gap 13, a first zone may be defined between the electronic display layer 14 and the cover panel 12, which may also be referred to as the LCD cavity, and a second zone may be defined between the electronic display layer 14 and the illumination device 16, which may also be referred to as the LED cavity. Any number of zones may be defined within the display assemblies 10.

(73) Operational ranges for the air circulating devices 48 may be established, such as at the controller(s) 52. Each of the operational ranges may be associated with one or more of the air circulating devices 48. Desired operating ranges may be established for the sensors 54, such as at the controller(s) 52. Each of the desired operating ranges may be associated with one or more of the sensors 54. Operational ranges for the air circulating devices 48 and/or desired operating ranges for the sensors 54 may be specific to the date, time, ambient conditions, combinations thereof, or the like, and may be programmed at, or stored at, the controller(s) 52. Operational ranges for the air circulation devices 48 and/or desired operating ranges for the sensors may be specific to the zone, air circulation device 48, and/or sensor 54 or for the whole display assembly 10. For example, without limitation, operational ranges for the air circulating devices 48 and/or desired operating ranges for the sensors 54 may be specific to day time or night time operations. Such day time and/or night time operations may be determined based on a location of the display assembly 10 and/or time of year (e.g., reflecting sunrise and/or sunset times based on location and date). In exemplary embodiments, without limitation, operational ranges for the air circulating devices 48 associated with the closed loop airflow pathway(s) and/or circulating gas 58 may be set to 100% fan speed during daytime hours and 15-100% during nighttime hours, and operational ranges for the air circulating devices 48 associated with the open loop airflow pathway(s) and/or ambient air 56 may be 0-100% at all times.

(74) Readings from the sensors 54 may be taken, such as continuously, periodically, sporadically, combinations thereof, or the like. Operation (e.g., speed, number of active fans, volumetric flow rate, power supplied, etc.) of some or all of the air circulation devices 48 may be controlled by a highest of readings from the sensors 54 at a given time or period of time relative to the associated desired operating range for the various sensors 54. For example, without limitation, the zone and/or sensor 54 having a highest reading relative to the maximum limit of the desired operating range associated with each zone or sensor 54 may be used by the controller(s) 52 to set the operating conditions of the air circulating devices 48 with the associated operational range. Such sensor 54 and/or zone may be controlling until subsequent readings indicate return to the desired operating range. Alternatively, or additionally, such sensor 54 and/or zone may be controlling until another sensor 54 and/or zone becomes highest and/or the furthers outside of the associated desired operating range. This may enhance thermal management by ensuring that the most problematic zone or sensor 54 reading is driving operations.

(75) The controller(s) 52 may be configured to ramp speed of the air circulating device(s) 48 up or down on a linear basis, inversely proportional ratio, by some multiple or other ratio relative to how

far the temperature is from the desired operating range, combinations thereof, or the like. Such adjustments may be made incrementally and readings retaken and adjustments made accordingly. (76) Such control may be performed on a zone-by-zone basis or for the entire display assembly **10**. For example, without limitation, air circulating devices **48** within a given zone and/or associate sensor(s) **54** may be adjusted individually based on such readings, or the air circulation devices **48** for the entire assembly **10** may be adjusted based on such readings, even if from a single zone and/or sensor **54**.

(77) Where a maximum operating temperature is reached or exceeded at one or more of the sensors **54**, operations of the air circulation devices **48** may be adjusted up to a maximum operational level (e.g., speed, number of active fans, volumetric flow rate, power supplied, etc.) within the operational ranges. If readings from the sensors **52** indicate that the maximum operating temperature subsequently remains reached or exceeded, power to the illumination device **16**, such as a backlight, may be reduced until temperatures fall below the maximum operating temperatures. Such reduction may be made in an inversely proportional fashion to how far the temperature has exceeded the maximum operating temperature. Such reduction may be made incrementally and adjustments made accordingly.

(78) Sensor readings may be continuously or periodically retaken and operations adjusted accordingly.

(79) Temperature readings from the sensors **54** may be communicated to the controller(s) **52** which may be configured to make operational determinations and adjustments for said air circulation devices **48** based on said readings. Alternatively, or additionally, such readings may be transmitted, such as by the network communication devices **62**, to one or more remote controller(s) **52** located remote from the display assembly **10**. Updates to the operational ranges for the air circulating devices **48** and/or desired operating ranges for the sensors **54** may be made from time to time, such as by way of instructions communicated to the controller(s) **52** through the network communication device **62** from one or more remote devices.

(80) The control logic shown and/or described with respect to the several figures and accompanying description provided herein may be used together or separately. For example, without limitation, the control logic shown and/or described with regard to FIG. **8** may be used with the condensation control logic of FIGS. **4-7B** or separately therefrom.

(81) While certain measures are shown and/or described herein in terms of degrees Celsius, equivalent measures in degrees Fahrenheit, Kelvin, or other measurement standards may be utilized.

(82) The ambient air **56** within the open loop airflow pathway(s) may be entirely or substantially prevented from mixing with the circulating gas **58** of the closed loop airflow pathway(s). For example, without limitation, the display assembly **10** may be configured to comply with various ingress protection standards, such as, but not limited to, IP **65**, IP **66**, IP **67**, IP **68**, combinations thereof, or the like, at least with regard to the closed loop airflow pathway(s) or other particular areas of the assembly **10**. Ambient air **56** may comprise air ingested from the surrounding environment and may or may not be filtered. The circulating gas **58** may comprise air kept fully or partially separate from the ambient air **56** in exemplary embodiments. For example, the circulating gas **58** may include ambient air **56** trapped when the assembly **10** is formed or otherwise periodically accessed (e.g., for servicing). Alternatively, or additionally, the circulating gas **58** may comprise filtered or purified air.

(83) Modified operations and/or condensation mitigation mode, which may be one and the same, may include establishing different temperature levels and/or offsets at which the fans or other thermal management components are activated and/or are operated. For example, without limitation, the fans or other thermal management components may be configured to operate relative to various temperature readings. The activation/deactivation points, operational speed and/or levels, combinations thereof, or the like of such fans or other thermal management components may be

varied between normal operating mode and modified operations and/or condensation mitigation mode. In exemplary embodiment, without limitation, fan speed or other operational characteristics of fans or other thermal management components may be set relative to temperature (e.g., at X temperature, Y fan speed, at Z temperature, A fan speed, etc.). Such relationship may be established by table, linear relationship, exponentiation relationship, logarithmic relationship, algorithmic relationship, combinations thereof, or the like. Certain preferred operating parameters, thresholds, or the like are disclosed herein which result in advantageous operation but are not necessarily intended to be limiting.

(84) Any embodiment of the present invention may include any of the features of the other embodiments of the present invention. The exemplary embodiments herein disclosed are not intended to be exhaustive or to unnecessarily limit the scope of the invention. The exemplary embodiments were chosen and described in order to explain the principles of the present invention so that others skilled in the art may practice the invention. Having shown and described exemplary embodiments of the present invention, those skilled in the art will realize that many variations and modifications may be made to the described invention. Many of those variations and modifications will provide the same result and fall within the spirit of the claimed invention. It is the intention, therefore, to limit the invention only as indicated by the scope of the claims.

(85) Certain operations described herein, may be performed by one or more electronic devices. Each electronic device may comprise one or more processors, electronic storage devices, executable software instructions, and the like, configured to perform the operations described herein. The electronic devices may be general purpose computers or specialized computing devices. The electronic devices may comprise personal computers, smartphones, tablets, databases, servers, or the like. The electronic connections and transmissions described herein may be accomplished by wired or wireless means. The computerized hardware, software, components, systems, steps, methods, and/or processes described herein may serve to improve the speed of the computerized hardware, software, systems, steps, methods, and/or processes described herein.

Claims

1. A display assembly providing condensation mitigation, said display assembly comprising: a housing; a cover layer forming a forward portion of the housing, wherein said cover layer is transparent or translucent; an electronic display located within the housing rearward of the cover layer, wherein said cover layer has an outer perimeter which extends beyond an outer perimeter of the electronic display; a continuous airflow pathway extending entirely within the housing such that the continuous airflow pathway is partitioned from an ambient environment, wherein the continuous airflow pathway extends, at least in part, forward and rearward of the electronic display to form a circuit; and one or more gaskets provided to, at least in part, partition said continuous airflow pathway from the ambient environment, wherein said one or more gaskets extend from an interior surface of the cover layer to a portion of the housing interior to said cover layer without contacting said electronic display so as to provide one or more gaps fluidly connecting portions of the continuous airflow pathway extending forward and rearward of the electronic display as part of said circuit.

2. The display assembly of claim 1 further comprising: one or more sensors located internal to said housing; and a controller in electronic communication with said one or more sensors and comprising software instructions stored at one or more non-transitory electronic storage devices, which when executed, configure one or more processors to: determine a dewpoint temperature for air at the display assembly; determine an internal temperature for the display assembly based, at least in part, on data received from said one or more sensors; calculate a dewpoint spread between said dewpoint temperature and said internal temperature; where said dewpoint spread is equal to or below a first predetermined value, initiate modified operations; and where said dewpoint spread is

equal to or above a second predetermined value, cease said modified operations.

3. The display assembly of claim 2 further comprising: an additional airflow pathway extending through at least a portion of said housing; and one or more fans located along said additional airflow pathway, wherein said modified operations comprise changing an operating speed for said one or more fans.
4. The display assembly of claim 3 wherein: said modified operations comprise reducing the operating speed for said one or more fans in accordance with a predetermined algorithm.
5. The display assembly of claim 3 further comprising: an intake and an exhaust located at said housing and fluidly connected to an ambient environment, wherein said additional airflow pathway comprises an open loop airflow pathway extending between said intake and said exhaust for ambient air; and one or more additional fans located along the additional airflow pathway, wherein said modified operations comprise modifying an operating speed for said one or more additional fans.
6. The display assembly of claim 5 wherein: said modified operations comprise reducing the operating speed for said one or more fans in accordance with a predetermined algorithm and increasing the operating speed for said one or more additional fans in accordance with the predetermined algorithm.
7. The display assembly of claim 2 wherein: said electronic display comprises one or more illumination devices; and said modified operations comprise increasing power supplied to the one or more illumination devices.
8. The display assembly of claim 2 further comprising: an additional airflow pathway extending through at least a portion of said housing; and said one or more sensors comprises a first sensor located within the additional airflow pathway.
9. The display assembly of claim 8 wherein: said additional airflow pathway is fluidly connected to an ambient environment; said one or more sensors comprises a second sensor located within the continuous airflow pathway; the data received from said second sensor is utilized by said controller to determine the internal temperature for the display assembly; and the data received from the first sensor is utilized by said controller to determine the dewpoint temperature.
10. The display assembly of claim 2 wherein: said one or more sensors comprise multiple temperature sensors; and the controller is configured to utilize data from a lowest reporting one of said multiple temperature sensors to determine the internal temperature.
11. The display assembly of claim 2 wherein: at least one of the one or more sensors comprises a humidity sensor; and the controller is configured to receive data from said humidity sensor to determine the dewpoint temperature.
12. The display assembly of claim 2 wherein: the controller is configured to utilize data from an internet-based weather reporting service to determine the dewpoint temperature.
13. The display assembly of claim 2 wherein: the controller is configured to receive data from one or more remote electronic device specifying or revising the first predetermined value and the second predetermined value.
14. The display assembly of claim 1 wherein: said electronic display comprises a liquid crystal display and light emitting diodes providing direct backlighting to the liquid crystal display; and the continuous airflow pathway is partitioned from the ambient environment in accordance with at least one of: ingress protection code **65**, **66**, **67**, and **68**.
15. The display assembly of claim 1 further comprising: additional software instructions stored at said one or more non-transitory electronic storage devices, which when executed, configure said one or more processors to: if a predetermined maximum internal temperature threshold is met or exceeded, reduce a power supplied to a backlight for the electronic display.
16. The display assembly of claim 1 wherein: each of the one or more gaskets comprise a rubberized material; and each of the one or more gaskets extend exclusively between the interior surface of the cover layer and the portion of the housing interior to said cover layer.

17. A display assembly providing condensation mitigation, said display assembly comprising: a housing comprising an intake and an exhaust; a cover layer forming a forward portion of the housing; an electronic display within the housing rearward of the cover layer; a first airflow pathway for ambient air extending within the housing between the intake and the exhaust; a second, continuous airflow pathway extending entirely within the housing such that the second airflow pathway is partitioned from an ambient environment; one or more sensors located internal to said housing; and one or more gaskets provided to, at least in part, partition said continuous airflow pathway from said first airflow pathway and the ambient environment, wherein said one or more gaskets extend from an interior surface of the cover layer to a portion of the housing.

18. The display assembly of claim 17 further comprising: a controller in electronic communication with said one or more sensors and comprising software instructions stored at one or more non-transitory electronic storage devices, which when executed, configure one or more processors to: determine a dewpoint temperature for air at the display assembly; determine an internal temperature for the display assembly based, at least in part, on data received from said one or more sensors; calculate a dewpoint spread between said dewpoint temperature and said internal temperature; where said dewpoint spread is equal to or below a first predetermined value, initiate modified operations; and where said dewpoint spread is equal to or above a second predetermined value, cease said modified operations.

19. A display assembly providing condensation mitigation, said display assembly comprising: a housing comprising a chassis or framework comprising metal; a cover layer forming a forward portion of the housing, wherein said cover layer is transparent or translucent and comprises at least one of: glass and a polymer; an electronic display for displaying electronic images located within the housing rearward of the cover layer, wherein said cover layer has an outer perimeter which extends beyond an outer perimeter of the electronic display such that the electronic images generated at the electronic display are visible through the cover layer; a continuous airflow pathway partitioned from an ambient environment and extending within the housing, including in front of the electronic display and behind the electronic display to form a circuit; at least one air circulation device within the continuous airflow pathway for moving gas through the continuous airflow pathway when activated; and one or more gaskets provided to, at least in part, partition said continuous airflow pathway from the ambient environment, wherein each of said one or more gaskets extend from an interior surface of a portion of the cover layer located beyond an outer perimeter of the electronic display to a portion of the chassis or framework, without contacting said electronic display, so as to provide gaps for the gas of the continuous airflow pathway along edges of the electronic display, thereby fluidly connecting portions of the continuous airflow pathway extending in front of, and behind, the electronic display as part of said circuit; wherein said continuous airflow pathway is partitioned from the ambient environment in accordance with at least one of: ingress protection code **65**, **66**, **67**, and **68**, at least in part by said one or more gaskets.

20. The display assembly of claim 19 wherein: each of the one or more gaskets comprise a rubberized material; and each of the one or more gaskets extends exclusively between the interior surface of the portion of the cover layer and the portion of the chassis or framework.
